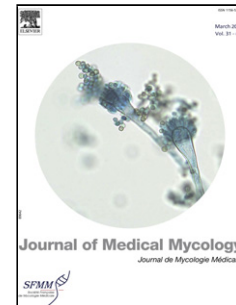


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Terbinafine Resistance in Dermatophytes: time to revisit alternate antifungal therapy

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Abstract

Resistant superficial dermatophytic infections of the skin and its appendages have emerged as a major health problem in India. Mutations in Squalene epoxidase gene have led to increasing incidence of resistance to terbinafine in dermatophytic isolates. We examined six patients with recalcitrant dermatophytosis attending Dermatology OPD at a tertiary care hospital and demonstrated terbinafine resistance by molecular method. Immediate hyperitivity (IH) reaction to Trichophyton antigen was highlighted in these patients. . The patients were treated with alternate antifungals after demonstration of resistance to terbinafine based on the antifungal susceptibility testing (AFST). On follow up the patients responded well to the substitute but the duration of therapy had to be prolonged six weeks of antifungal treatment.

Keywords: Tinea corporis/cruris; Terbinafine; *Trichophyton interdigitale*; SQLE mutation

Introduction

Dermatophytes are pathogenic fungi that have a high affinity for the keratin of skin, hair and nail. With the estimated global prevalence of 20-25%, the chronic nature of dermatophytosis is gaining importance because of the failure of treatment or perhaps due to the acquisition of drug resistance. The epidemiological profile of dermatophytosis has demonstrated an increase in the diversity with the dynamic evolution of dermatophytes aided by the migration of population (1, 2). Recently an increase in the number of cases of recalcitrant dermatophytic infections of skin has reportedly caused a high degree of concern amongst clinicians in India. This changing trend in the clinical profile of patients manifesting predominantly as tinea corporis or cruris has been attributed to various factors. Terbinafine is still considered the drug of choice for dermatophytosis however resistance to terbinafine remains largely undiagnosed due to non-availability of molecular tools. The continued use of terbinafine in patients with recalcitrant dermatophyte infections may lead to the emergence of mutations and selection of these species responsible for the progression to the chronicity of infection. Therefore the importance of detecting drug resistance in clinical practice and accordingly apply modified strategy in treating such cases would be beneficial to the patient. Further taxonomic controversies have led to confusion in renaming the agent, though Chowdhary et al. have identified that *T mentagrophyte*/ *T. interdigitale* require reassessment as per the NCBI database available; however, the nomenclature of *T. mentagrophyte*

complex is acceptable till the issues are resolved (3).

Additionally to the emergence of resistance to antifungals, topical steroid misuse, suboptimal dosing and treatment a dysregulated host immune response can also be an underlying factor in recalcitrant dermatophytosis(4,5,6). The molecular mechanisms of drug resistance in dermatophytes involve mutations in genes encoding for drug target enzymes (6) or disrupting binding and interaction of antifungal agents to fungal enzyme molecules as demonstrated in reports. Popular among the antifungals, terbinafine (TRB), has been used orally and topically as a first-line drug in the therapy of dermatophyte infection for the past several years. Documented molecular mechanism of resistance to TRB is presently limited to clinical *T. rubrum* and *T. interdigitale* strains where the resistance has been predominately attributed to point mutations in the SQLE target gene resulting in a single amino acid substitution at one of four positions (Leu393, Phe397, Phe415, His440) of the SQLE protein. However, there appears to be an unpredictable association between the MIC values and the clinical outcome, in patients with dermatophytic infections (6, 7, 8). The increasing incidence of chronic and recalcitrant dermatophytic infections are now considered as an important public health problem, constituting a large bulk of cases attending dermatology OPD. With a very limited armamentarium of antifungals effective against dermatophytes and recent studies showing resistance to antifungals (especially to terbinafine), the in vitro antifungal susceptibility testing must be adopted by laboratories to monitor the emerging resistance in the community. The situation also demands robust research in the pathogenesis of dermatophytosis and its interaction with the host immunity and to discover new targets for drug development. Here we examined six patients with recalcitrant dermatophytic infections attending Dermatology OPD at a tertiary care hospital and investigated the fungal and immunological profile at the time of recruitment and recorded the clinical outcome at six weeks of treatment. The fungal isolates were subjected to SQLE mutation analysis by conventional PCR and correlated with the patient's clinical outcome.

Cases

Six patients of chronic dermatophytosis attending Dermatology OPD of a tertiary care hospital, who failed to respond to treatment with terbinafine were followed up after initiating alternate therapy with other antifungals. Clinical assessment was done after 2.5 months of treatment. The demographic details of the patients are depicted in Table 1.

Mycological identifications

The clinicians (Dermatologist) at the hospital diagnosed the patients with recalcitrant tinea infection as no or minimal response to treatment after 6 weeks of either oral and/or topical antifungal use confirmed by KOH mount of skin scrapings. Detailed clinical examination was done and recorded.

The specimen (skin scraping) were collected using a sterile scalpel on a black paper after disinfection with 70% alcohol before collecting the scraping. The sample was taken from the advancing edge of lesions by scraping from the centre to edge crossing the margins using a sterile scalpel. Samples were divided into two portions for microscopy and culture. A portion of the scrapings was subjected to 10% Potassium hydroxide (KOH) digestion followed by direct microscopy while the rest of scrapings were inoculated on SDA (Sabouraud dextrose agar) with an antibiotic (0.05 mg/ml chloramphenicol and 0.5 mg/ml cycloheximide) and incubated at 28 °C. Growth on SDA tubes was observed weekly for four weeks and

further processing of growth was done to identify the isolates. The identification of dermatophyte was done by conventional phenotypic and genotypic methods.

Mycological identifications

The culture isolates on SDA were subjected to DNA extraction using a commercially available DNA extraction kit (HiYield Genomic DNA Kit; Real Biotech Corporation, Taiwan). Genomic DNA extracted was subjected to polymerase chain reaction (PCR) using pan fungal primers; Internal Transcribed Spacer (ITS)-1 (5'-TCCGTAGGTGAACCTGCGG-3') and ITS-4 (3'-TCCTCCGCTTATTGATATGC-5') region of 18s rRNA. PCR amplicons were subjected to purification and DNA sequencing. The DNA sequencing was performed for ITS1 and ITS4 region commercially (Xcelris Labs Limited, India). The sequences were analysed and compared with the sequences deposited in GenBank by using the BLAST (<http://www.ncbi.nlm.nih.gov/BLAST/Blast.cgi>). The ITS sequences showed ≥99% similarity with *Trichophyton interdigitale*. The 18S rRNA sequences of all the six isolates were submitted to GenBank and were assigned accession numbers (MN010328 to MN010333).

Molecular identification squalene epoxidase (SQLE) gene in *T. interdigitale* isolates

The amplification for SQLE gene was performed using primers, TrSQLE-F1, 5'-ATGGTTGTAGAGGCTCCTCCC-3' TrSQLE-R1, 5' -CTAGCTTTGAAGTTCGGCAAA -3' (9, 10). The samples were amplified by using the following thermal cycling parameters, initial denaturation at 95°C for 5 min, 35 cycles of amplification consisting of denaturation at 95° C for 30 sec, annealing at 60°C for 45sec and extension at 72 °C for 30 sec followed by a final extension at 72 °C for 10 min. Purification and sequencing of the PCR products were done commercially. The sequences obtained were compared with reference Genbank sequence of *T. interdigitale*. All the six isolates were identified for the mutation in SQLE gene (Table 2).

Antifungal susceptibility testing

Antifungal susceptibility testing (AFST) was performed as per CLSI-BMD, M38-A2 guidelines (10). The antifungals tested included fluconazole (FLU; Sigma-Aldrich, USA), itraconazole (ITR; Sigma-Aldrich, USA), terbinafine (TRB Sigma-Aldrich, USA) and griseofulvin (GRE; Sigma-Aldrich, USA). The isolates were grown on SDA for 7 -10 days and conidia were scrapped to prepare inoculum in saline with Tween 80. The final inoculum was adjusted to 1×10^3 to 3×10^3 CFU/ml counted by haemocytometer. The MICs of quality-control ATCC strains of *T. rubrum* ATCC 28188, *Candida parapsilosis* ATCC 22019 and *Candida krusei* ATCC6258 along with drug and mould free controls were included and microtitre plates were incubated at 35oC . Minimum inhibitory concentration endpoints for all the drugs were defined as the lowest concentration that inhibited 100% of growth compared to growth control wells for itraconazole, terbinafine, griseofulvin, as reading visually, at 72 hrs. For fluconazole, a 50 % reduction in growth as compared to growth control was considered as the MIC range for QC strain of *T mentagrophyte* ATCC MYA 4439 against griseofulvin, itraconazole, voriconazole and terbinafine were interpreted as per Clinical Laboratory Standard Institute (CLSI)M 38A2 E. The MIC range for QC strain ATCC MYA 4439 against fluconazole, ketoconazole, sertaconazole, luliconazole, clotrimazole has not been given by CLSI. For dermatophytes, no clinical breakpoints (CBPs) for MIC values have yet been established to guide antifungal therapy. In the absence of these, epidemiological cut-off values (ECVs or ECOFFs) may serve as a temporary substitute for the detection of resistance until the process of complicated CBP determination

has finished. The ECOFFs may be helpful for physicians in the current scenario of recalcitrant dermatophytoses in India until CBPs based on clinical outcome data are established.

Trichophytin skin testing

Trichophytin intradermal skin testing was performed on the patients and the test result read within 15 min and after 48 h. Trichophytin antigen (with protein concentration 2.3 mg/ml) derived from anthropophilic *Trichophyton spp* strains was procured from All Cure Pharma Pvt. Ltd., Delhi (India); 0.01 ml each of trichophytin antigen and phosphate-buffered saline (PBS) (negative control) was injected intradermally using separate syringes at independent sites of the skin of the forearm in all patients. The test was read after 15 min for immediate hypersensitivity (IH) response and 48 h for delayed-type hypersensitivity (DTH) response. Any reaction with mean wheal diameter at least 3 mm greater than the negative control was taken as positive (10).

Result

The six patients included in this study were in the age group of 22-45 year age group. They were either diagnosed with Tinea cruris (1) or Tinea corporis (2) or a combination of two (3). The duration of the disease varied from 3 months to 12-24 months. The Trichophytin skin intradermal testing showed immediate hypersensitivity(IH) reaction in all the cases. Delayed hypersensitivity reaction was not exhibited

All had previously received a course of terbinafine therapy for the varying period. Griseofulvin was given 20mg/ kg body weight twice daily for 4 months sequentially followed by itraconazole 100mg twice daily for 2 months as an alternate therapy. After 2.5 months of modified treatment, all reported clinically cured(Table 1).

The in-vitro antifungal susceptibility profiles of all six patients are summarised in Table 2, indicated high MIC to TRB with a value $\geq 1\mu\text{g/ml}$ in five(83%) isolates, while one showed low MIC value $\leq 0.5 \mu\text{g/ml}$ for TRB. The mutation analysis for SQLE indicated three of six isolates, with amino acid substitution at Phe397Leu (F397L) while one showed amino acid substitution at two positions Phe397Leu and Tyr394Asn (Y394N). High MIC for Fluconazole was also observed in four (66.6%) isolates. All isolates showed low MIC to griseofulvin.

Discussion

The epidemiological profile of dermatophytosis has demonstrated an alarming change due to the chronic nature of the infection. Recalcitrant tinea is defined as poor response to adequate dose and duration of treatment with antifungals, confirmed by positive microscopy and culture. The chronicity has been attributed to the irrational use of antifungals and misuse of topical steroid preparations. In our study, the patients belonged to adult working populations engaging in outdoor activities in the hot and humid climate. This environment favours the persistence and continued survival of dermatophytes on the skin and its appendages. Lifestyle changes, environmental factors have promoted a shift in the spectrum of this fungal biota towards an anthropophilic adaptation (2).

Several virulence factors are excreted by dermatophyte during its invasion into the host tissue which induces host immune responses. Antigens derived from *Trichophyton* exhibit unusual immunologic

properties based on their ability to induce distinct skin test reactions in different individuals. IH skin tests to *Trichophyton* extract are immunoglobulin E (IgE) antibody (Ab)-mediated reactions which are frequently observed in subjects with chronic dermatophytosis (11). The skin response to Trichophyton antigen can be considered as a marker of immune status in fungal infections and aid as a prognostic indicator of treatment outcome.

The Indian isolates have experimentally shown to secrete high levels of proteases particularly subtilisins as the most abundant enzyme belonging to endoprotease family, suggesting their preferential expression under different levels of stress (like antifungals and high MIC, etc.). REF IGIB Drug resistance in dermatophytes has been reported from a few centres across the country, though the undefined breakpoints as per CLSI remains a setback on its association with clinical outcome (12). Nevertheless, laboratories must generate MICs to various antifungals used in clinical practice against the dermatophytes so that the epidemiological cut-offs can be generated for better association studies. Currently, Fluconazole resistance is at 40%, terbinafine 10-20%, Itraconazole at 2%, griseofulvin at 40% and voriconazole, sertaconazole, luliconazole remain susceptible as reported in India(5).

In our study, we observed high MIC to terbinafine in all the 6 cases and 3 cases showed point mutation at F397L identified by PCR and sequencing. Such isolates exhibiting point mutations at F397 L prove to be non-responders to all squalene epoxidase inhibitors (10). On clinical failure at 6 weeks of treatment, patients were put on different antifungal regimens, and all showed improvement. Prolonged exposure to terbinafine can favour the selection of *Trichophyton spp.* resistant strains and can be the source of an emerging health burden. On the other hand, patients with no mutations at SQLE, respond to conventional-dose or high dose of terbinafine. In our study, the clinicians decided to treat these patients (case no. 2,4 and 5) with Tab griseofulvin and or itraconazole. Terbinafine MIC₉₀ <0.06ug/ml concentration kills the fungus in a few days; Resistance increases in sequential samples after exposure to the drug, though ERG 1 mutation may not always cause treatment failure. Similarly, constant use of azoles may give rise to poor susceptibility to the drugs(13, 14). The constant use of azoles may lead to a rise in MIC values in Trichophyton though further studies are needed to co-relate the MIC values with clinical outcome in these patients. Ebert et al documented in a large multicentric study that a strong, positive, linear correlation between MICs of itraconazole and voriconazole was demonstrated. (15). Isolates with reduced susceptibility to itraconazole tended to be less susceptible to voriconazole, and vice versa. These findings may implicate that triazole resistance in dermatophytes is not drug-specific but is mediated by a shared mechanism of resistance. We observed that reduced susceptibility to triazoles in Trichophyton isolates was associated with the amino acid substitution Ala448Thr at the C-terminus of the squalene epoxidase. On average, strains containing an Ala448Thr substitution exhibited higher itraconazole and voriconazole MICs. However, there are strains with the Ala448Thr substitution that showed low sensitivity to azoles.(15)Consequently, this mutation alone cannot be incriminated for a lack of sensitivity to azoles, though it could be interpreted that such mutations influence later steps of ergosterol synthesis through conformational changes and result in reduced susceptibility to azoles. For dermatophytes, since no clinical breakpoints (CBPs) for MIC values have yet been established to guide antifungal therapy., the epidemiological cut-off values (ECVs or ECOFFs) may serve as a temporary substitute for the detection of resistance until the process of complicated CBP determination has finished. These ECOFFs may be helpful

for physicians in the current scenario of recalcitrant dermatophytosis in India until CBPs based on clinical outcome data are established(15).

In our study, four isolates showed F397L mutation correlated with high MIC values. Two patients who did not show any mutation in the isolates had low MIC values to terbinafine. According to Khurana et al. such isolates with high MIC but no mutation can be overcome by the higher dosage of terbinafine or patients who have not been previously exposed to terbinafine may benefit with a higher dosage of the drug(16). Hence optimizing drug exposures can improve clinical outcome. In our study *T mentagrophyte* showing MIC range 2-8 ug/ml did not respond to terbinafine but responded to alternate therapy like griseofulvin or itraconazole. Consequently, the isolates did not show any cross-resistance to fluconazole or griseofulvin. It is also difficult to say whether the resistance was primary or acquired therefore long term clinical follow up with in-vitro antifungal susceptibility monitoring on a larger number of patients may be necessary. The absence of clinical cure in patients on terbinafine was considered in absence of reduction in number and size of lesions, the persistence of itching and raised intensity of inflammation observed at the end of 4- 6 weeks. Most dermatophytes acquire resistance from a population previously sensitive to the drug. Constant use and selective pressure can often lead to low frequency gene mutation. The resistant strains become predominant unless the mutations that rendered the fungus-resistant also reduces their adaptability and minimizes its chance of survival in the presence of the drug which may eventually lead to the loss of the species. Another reason could be the sub MIC concentration of drug promoting adaptation of this fungus to evade stress and enhance the production of proteases for their survival. Though SQLE mutations appear as the most plausible reason for the poor clinical outcomes in these patients, the underlying altered host immune response may have a role in the poor clearance of the fungi as was observed in these patients exhibiting a poor DTH response to Tricophyton skin antigen test.

T. interdigitale has emerged as the predominant pathogen associated with recalcitrant dermatophytosis. Steroid modified clinical pattern of dermatophytosis has emerged as a challenge to clinicians to choose the correct management. There is an urgent need to rationalise the use of both topical and systemic antifungals to maximize the clinical effect and achieve clinical cure. The high MIC to terbinafine indicates its failed response to eliminate the fungus; hence clinicians need to consider alternate drugs in recalcitrant Tinea. The possibility of increasing terbinafine resistance and reduce squalene epoxidase activity as underlying key factors in poor clinical cure cannot be ignored. However further research to evaluate the interplay of virulence factors of these organisms and their contribution to chronicity and recalcitrance of the infections will be a valuable contribution (17, 18, 19).

We conclude stating that high terbinafine resistance in India has been documented. It may no longer be the drug of choice for T corporis. Clinicians should not give a low dosage of antifungals and create sub MIC levels for drug resistance to develop. It is suggested that terbinafine resistance may be monitored with a rapid test to detect SQLE gene mutations. Thereafter start with a different class of antifungals if terbinafine resistance is reported. Wait for the mycological cure and not stop therapy until the clinical reduction in the lesion is observed.

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Conflict of Interest

There are none.

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Figure Caption

Cases	Identification (Sequencing)	Mutation (SQLE)	Fluconazole	Itraconazole	Terbinafine	Griseofulvin
1.	<i>T.mentagrophytes</i> <i>Interdigitale</i>	F397L	2	0.0625	2	0.0625
2.	<i>T.mentagrophytes</i>	F397L	0.125	0.0313	4	0.125
3.	<i>T.mentagrophytes</i>	F397L	4	0.125	2	1
4.	<i>T.mentagrophytes</i>	--	128	0.0313	2	0.125
5.	<i>T.mentagrophytes</i>	--	1	0.5	0.125	0.5
6.	<i>T.mentagrophytes</i>	F397L, Y394N	64	2	8	0.125

Table 2. MIC profile of the dermatophyte isolates with SQLE mutation